

every output inside the rigorous error envelope (all policies) 6 proof-carrying code Ed25519-signed certificate; quorum by co-signature Rungs 35 execute deterministic probe inputs (seeded from the program digest) at probe extents; bounds are established at the maximal extents, which the monotonicity lemmas (admissible mono , withinEnvelope mono) show covers every smaller extent.

Zero-knowledge proofs of integer inference: a quantized network compiled to a rank-1 constraint system (R1CS) over a prime field, a witness that satisfies it, and both written in the formats of circom/snarkjs (.r1cs , .wtns) so any Groth16 or PLONK prover of that ecosystem can prove, and any verifier (including the Solidity one snarkjs exports) can check, that this public model mapped a private

input to a public output. model = :linear, w1, b1 , :relu, :linear, w2, b2 w1 int8, weights public c = Vapor.ZK.compile(model, k: 16) :ok, wit = Vapor.ZK.witness(c, x) x : int8, kept private :ok = Vapor.ZK.check(c, wit) Vapor.ZK.write r1cs(c, m.r1cs); Vapor.ZK.write wtns(c, wit, m.wtns) Why the integer fragment, and why public weights Field arithmetic is integer arithmetic. + and of a field are exactly those of the integers as